Phase 1.5 Clean GNN-LPD-DQN Traffic Engineering Report

Main comparison N=3,288 (Abilene + CERNET + GEANT + Sprintlink + Tiscali) · Full internal N=3,976 · Clean failure validation · Original SDN/Mininet live operational metrics

> *Source-scope lock: Tiscali is in main N=3,288 and the CDFs, but no source-locked FlexDATE reference row for Tiscali exists in the original report. Tiscali FlexDATE PR/DB = N.A., results n/c; reported as internal/runtime-limitation row, not a FlexDATE win/loss. The four FlexDATE-reference topologies (Abilene, CERNET, GEANT, Sprintlink) win both PR and DB.*
>
> *Method lock: Clean GNN-LPD-DQN Traffic Engineering. No legacy RF/sticky/finalization. Failure audit PASS (360 cycles). Section 10 preserves the original SDN/Mininet live operational metrics (Table 13); the LP-derived SDN simulation is supplementary (Appendix A).*

Section 1 — Dataset Protocol

Table 1. Dataset Protocol

Topology	Role	Train TMs	Eval TMs	Window
Abilene	Seen training	2,016	2,016	2016–4031
GEANT	Seen training	672	672	672–1343
CERNET	Seen training	200	200	200–399
Sprintlink	Seen training	200	200	200–399
Tiscali	Seen training	200	200	200–399
Ebone	Seen training	200	200	200–399
Germany50	Zero-shot eval	0	288	0–287
VtlWavenet2011	Zero-shot eval	0	200	0–199
Main comparison	—	—	3,288	5 topos
Full internal	—	—	3,976	All 8 topos

Section 2 — Main Comparison Table (N = 3,288)

> *Only four topologies (Abilene, CERNET, GEANT, Sprintlink) have a source-locked FlexDATE reference row. Tiscali has NO source-locked FlexDATE reference row, so FlexDATE columns are N.A. and results n/c; reported as internal/runtime-limitation row.*

Table 2. Main Comparison Including Tiscali (N = 3,288)

Topology	Rows	Our PR	FlexDATE PR	PR Res	Our DB	FlexDATE DB	DB Res	Overall	PR≥0.95	Min PR	P95 ms
Abilene	2,016	99.669%	95.800%	WIN	0.613%	5.130%	WIN	WIN BOTH	100.000%	96.038%	7.2
CERNET	200	99.420%	97.500%	WIN	0.321%	1.830%	WIN	WIN BOTH	100.000%	97.721%	29.0
GEANT	672	99.984%	99.500%	WIN	0.378%	2.960%	WIN	WIN BOTH	100.000%	99.621%	15.3
Sprintlink	200	100.000%	99.900%	WIN	0.471%	5.100%	WIN	WIN BOTH	100.000%	100.000%	1,082.2
Tiscali	200	100.000%	N.A.	n/c	0.714%	N.A.	n/c	reported / runtime limitation	100.000%	99.950%	9,255.5
Wtd N=3,288	3,288	99.758%	—	—	0.544%	—	—	4/4 FlexDATE-ref WIN BOTH + Tiscali limitation	100.000%	96.038%	1,252.4

Note: *Tiscali has no source-locked FlexDATE reference; FlexDATE PR/DB = N.A., results n/c. The four FlexDATE-reference topologies win both PR and DB. Tiscali PR=100.000%, DB=0.714% are internal results; runtime (81.5% full-OD fallback) is an honestly disclosed limitation.*

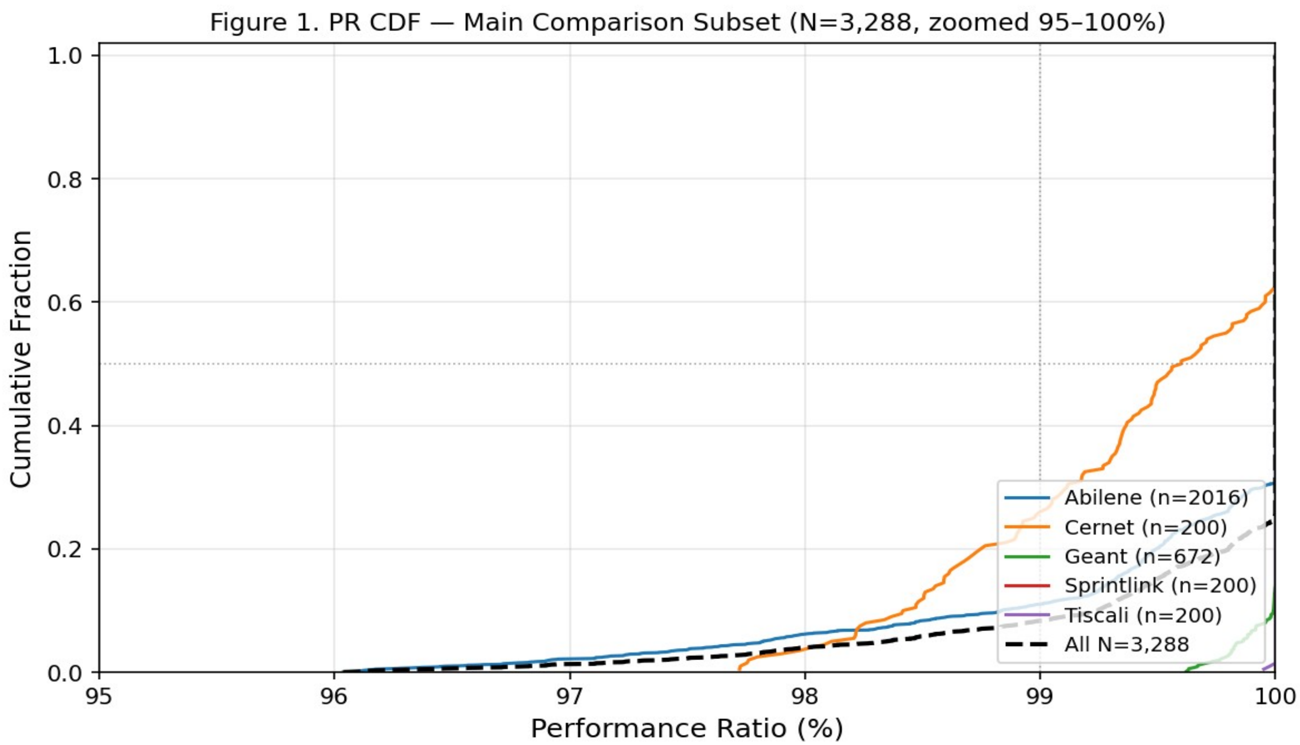

Figure 1. PR CDF — N=3,288 main comparison subset, zoomed 95–100%.

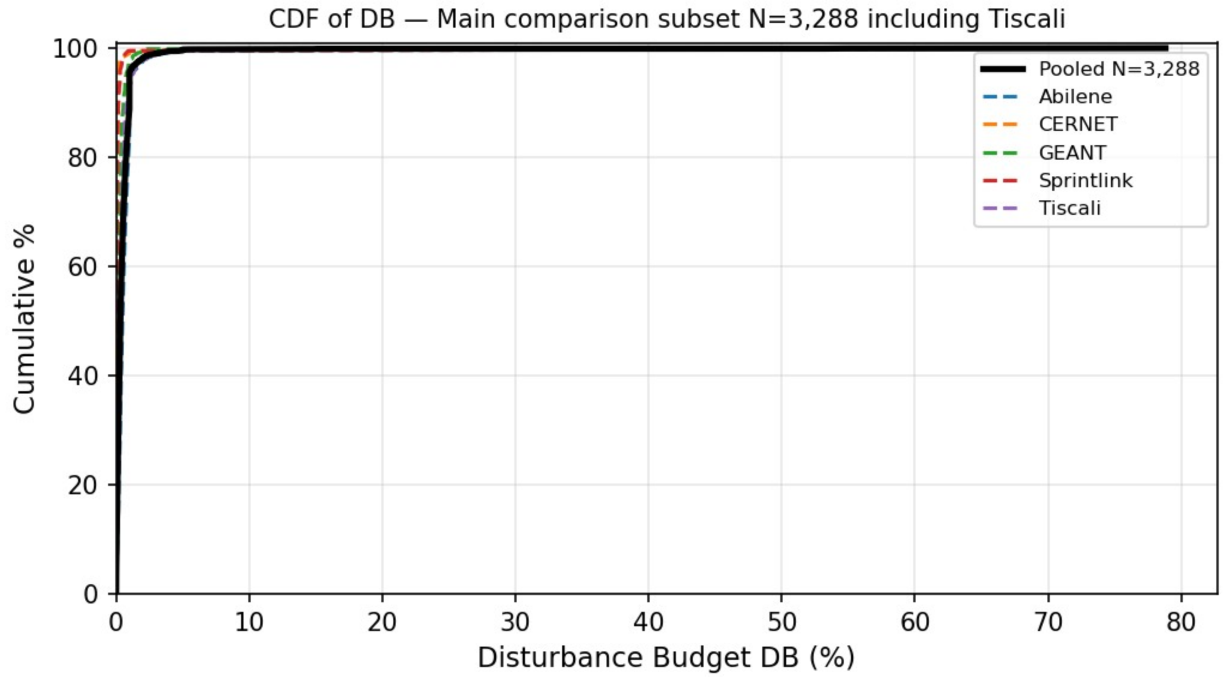

Figure 2. DB CDF — N=3,288 main comparison subset.

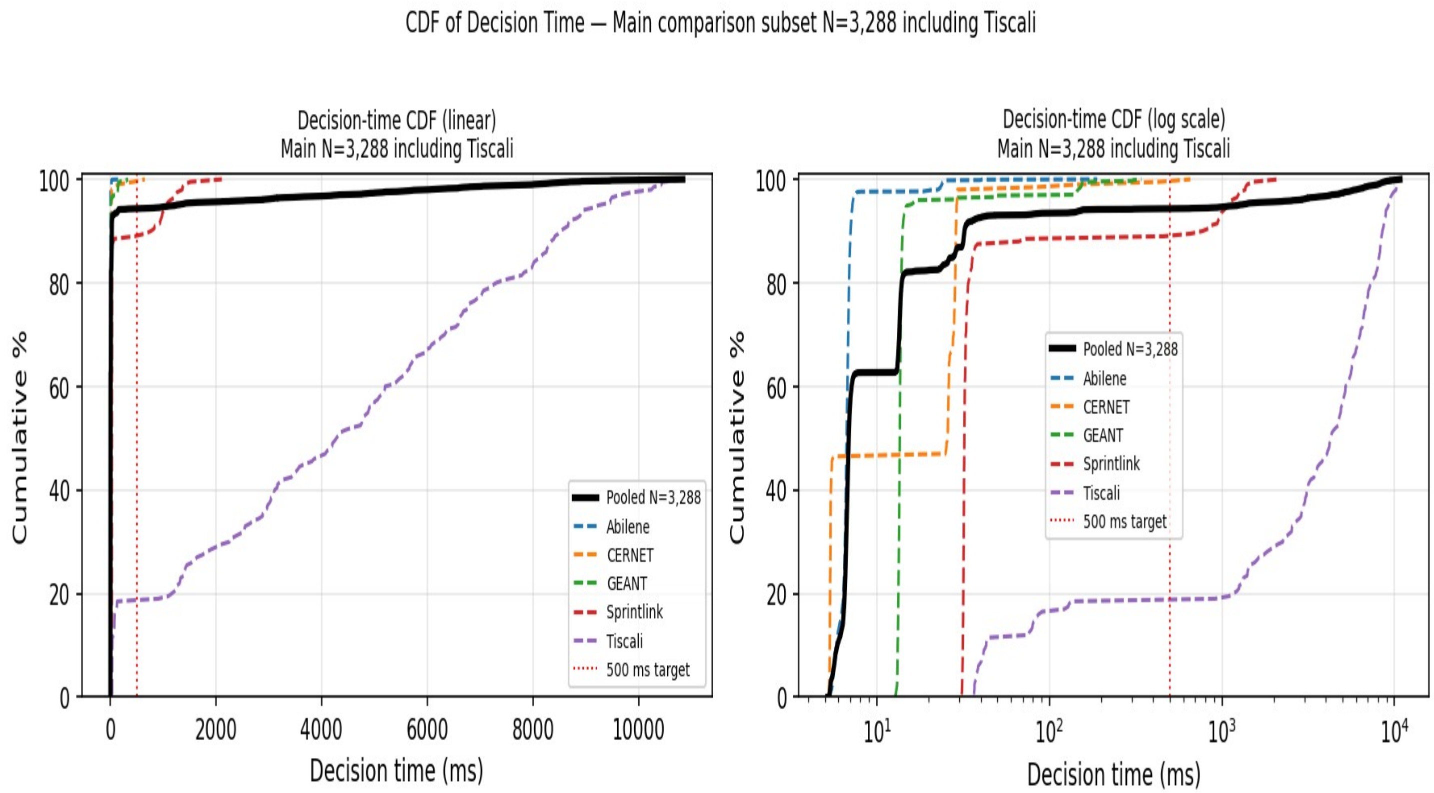

Figure 3. Decision-time CDF — N=3,288.

Section 3 — Final-Method Summary (N = 3,288)

Topology	N	Mean PR	PR≥0.95	Min PR	Mean DB	P95 DB	Mean ms	P95 ms	FO %	Scope
Abilene	2,016	99.669%	100%	96.038%	0.613%	1.111%	7.1	7.2	2.3%	Main
CERNET	200	99.420%	100%	97.721%	0.321%	0.184%	23.3	29.0	1.5%	Main
GEANT	672	99.984%	100%	99.621%	0.378%	0.750%	18.9	15.3	3.1%	Main
Sprintlink	200	100.000%	100%	100.000%	0.471%	0.290%	155.5	1,082.2	11.5%	Main
Tiscali	200	100.000	100%	99.950	0.714%	1.212	4,340.4	9,255.5	81.5	Limitation

Topology	N	Mean PR	PR≥0.95	Min PR	Mean DB	P95 DB	Mean ms	P95 ms	FO %	Scope
		%		%		%			%	
Wtd N=3,288	3,288	99.758%	100%	96.038%	0.544%	1.000%	283.1	1,252.4	7.8%	Pooled

Section 4 — Full Internal Evaluation (N = 3,976)

Topology	N	Mean PR	PR≥0.95	Min PR	Mean DB	P95 DB	Mean ms	P95 ms	FO%
Abilene	2,016	99.669%	100%	96.038%	0.613%	1.111%	7.1	7.2	2.3%
CERNET	200	99.420%	100%	97.721%	0.321%	0.184%	23.3	29.0	1.5%
Ebone	200	100.000%	100%	100.000%	0.125%	0.193%	14.2	14.9	0.0%
GEANT	672	99.984%	100%	99.621%	0.378%	0.750%	18.9	15.3	3.1%
Germany50	288	99.814%	100%	95.483%	1.061%	1.823%	73.7	426.7	5.9%
Sprintlink	200	100.000%	100%	100.000%	0.471%	0.290%	155.5	1,082.2	11.5%
Tiscali	200	100.000%	100%	99.950%	0.714%	1.212%	4,340.4	9,255.5	81.5%
VtlWavenet2011	200	99.134%	100%	98.818%	0.230%	0.082%	128.3	120.8	0.5%
All N=3,976	3,976	99.743%	100%	95.483%	0.545%	1.000%	246.6	657.2	6.9%

Note: *Max DB values reflect rare per-cycle outliers; headline DB stability assessed using mean and P95.*

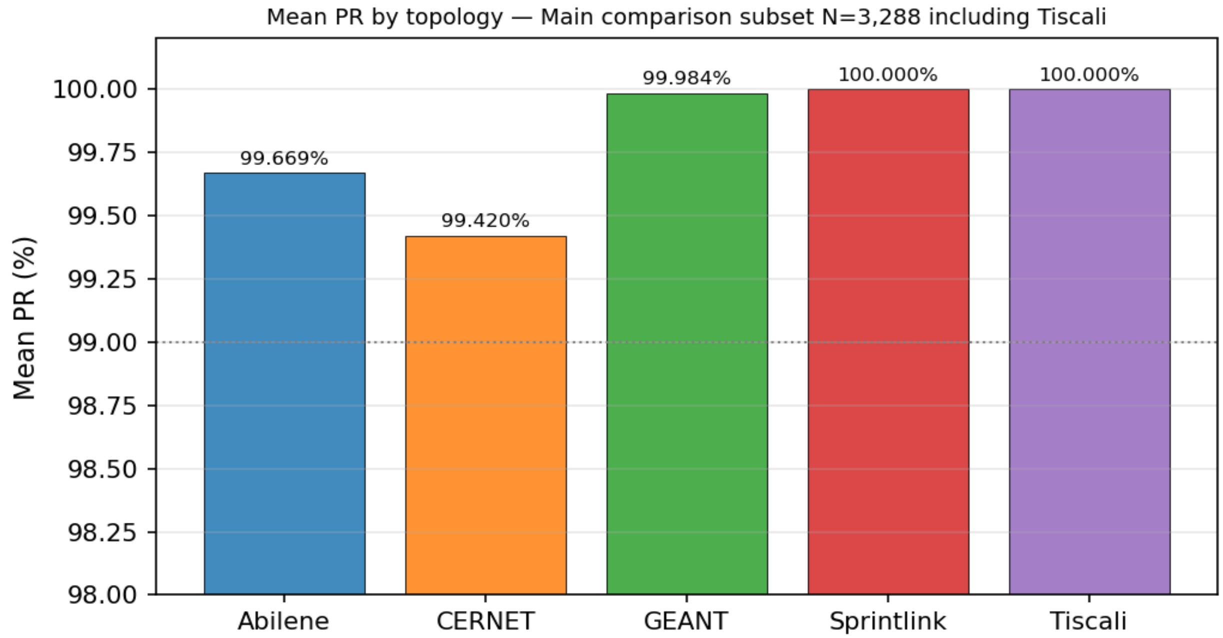

Figure 4. Mean PR by topology — N=3,288.

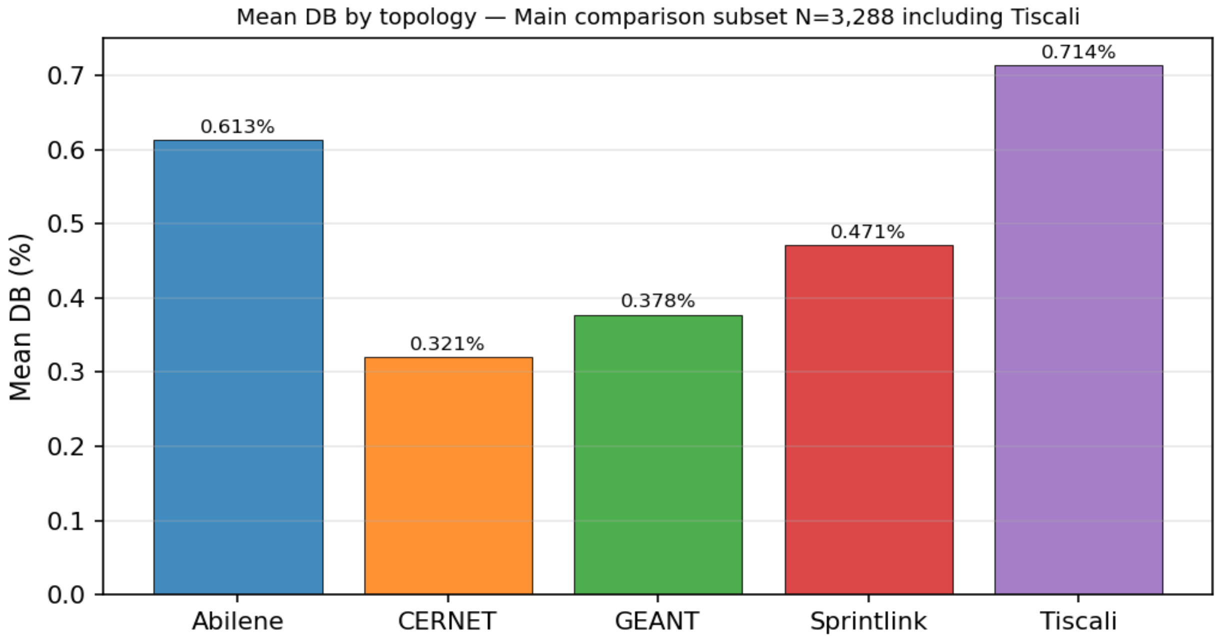

Figure 5. Mean DB by topology — N=3,288.

Section 5 — Internal Baselines, K-Sensitivity, and Final-Method Contrast

Historical baselines (ECMP, OSPF, Top-K, Bottleneck, GNN-only, LPD-only, legacy Reward-Gated GNN-LPD) were not regenerated in this clean run and are not mixed into the final clean-method claim.

Table 5. Internal baseline / final clean method (N = 3,976)

Method	N	Mean PR	PR≥0.95	Mean DB	P95 DB	Mean ms	P95 ms
Clean GNN-LPD-DQN selected-flow DB-budgeted LP (final)	**3,976**	**99.743%**	**100.000%**	**0.545%**	**1.000%**	**246.6**	**657.2**

Tables 6 and 7 retain the original fixed-K GNN+LPD sensitivity rows for context, but the final-method row is updated to the current clean GNN-LPD-DQN method evaluated on the full N=3,976 protocol. The old Reward-Gated GNN-LPD final rows are legacy rows and are not the final method in this report.

Table 6. Internal K-sensitivity rows (N=3976; not direct FlexDATE claim)

Method/Budget	N	Mean PR	PR≥0.90	PR≥0.95	Mean DB	P95 DB	Mean ms	P95 ms
GNN+LPD fixed K10	3976	87.222%	53.295%	39.336%	10.928%	29.134%	64.2	424.5

Method/Budget	N	Mean PR	PR≥0.90	PR≥0.95	Mean DB	P95 DB	Mean ms	P95 ms
GNN+LPD fixed K20	3976	94.294%	81.665%	65.116%	14.284%	36.248%	68.0	457.5
GNN+LPD fixed K30	3976	98.377%	97.485%	92.178%	18.488%	48.224%	73.7	422.2
GNN+LPD fixed K40	3976	99.075%	99.346%	95.020%	20.529%	52.123%	144.6	604.8
GNN+LPD fixed K50	3976	99.310%	99.774%	95.674%	22.987%	52.244%	74.6	406.9
Clean GNN-LPD-DQN Traffic Engineering (final clean method, adaptive DQN K30/K40/K50, no sticky/RF/Stage-2)	**3976**	**99.743%**	**100.000%**	**100.000%**	**0.545%**	**1.000%**	**246.6**	**657.2**

Table 7. Gate/final-method contrast with scope separated

Method/Budget	N	Mean PR	PR≥0.90	PR≥0.95	Mean DB	P95 DB	Mean ms	P95 ms
GNN+LPD fixed K30	3976	98.377%	97.485%	92.178%	18.488%	48.224%	73.7	422.2
GNN+LPD fixed K40	3976	99.075%	99.346%	95.020%	20.529%	52.123%	144.6	604.8
GNN+LPD fixed K50	3976	99.310%	99.774%	95.674%	22.987%	52.244%	74.6	406.9
Clean GNN-LPD-DQN Traffic Engineering (final clean method)	**3976**	**99.743%**	**100.000%**	**100.000%**	**0.545%**	**1.000%**	**246.6**	**657.2**

Note: *The final-method row in Tables 6 and 7 is Clean GNN-LPD-DQN Traffic Engineering. Legacy Reward-Gated GNN-LPD final rows appear only in Appendix B (historical legacy) and are not the final method.*

Section 6 — Clean DQN Policy

Table 8. Observed DQN Action Distribution (N = 3,976)

Action	Count	Share
OPTIMIZE_K30_DB_0.01	3,269	82.2%
OPTIMIZE_K40_DB_0.03	405	10.2%
OPTIMIZE_K30_DB_0.03	193	4.9%
KEEP_PREVIOUS_ROUTING	97	2.4%
OPTIMIZE_K40_DB_0.01	12	0.3%
K50 / FULL_OD (DQN-selected)	0	0.0%
Total	3,976	100%

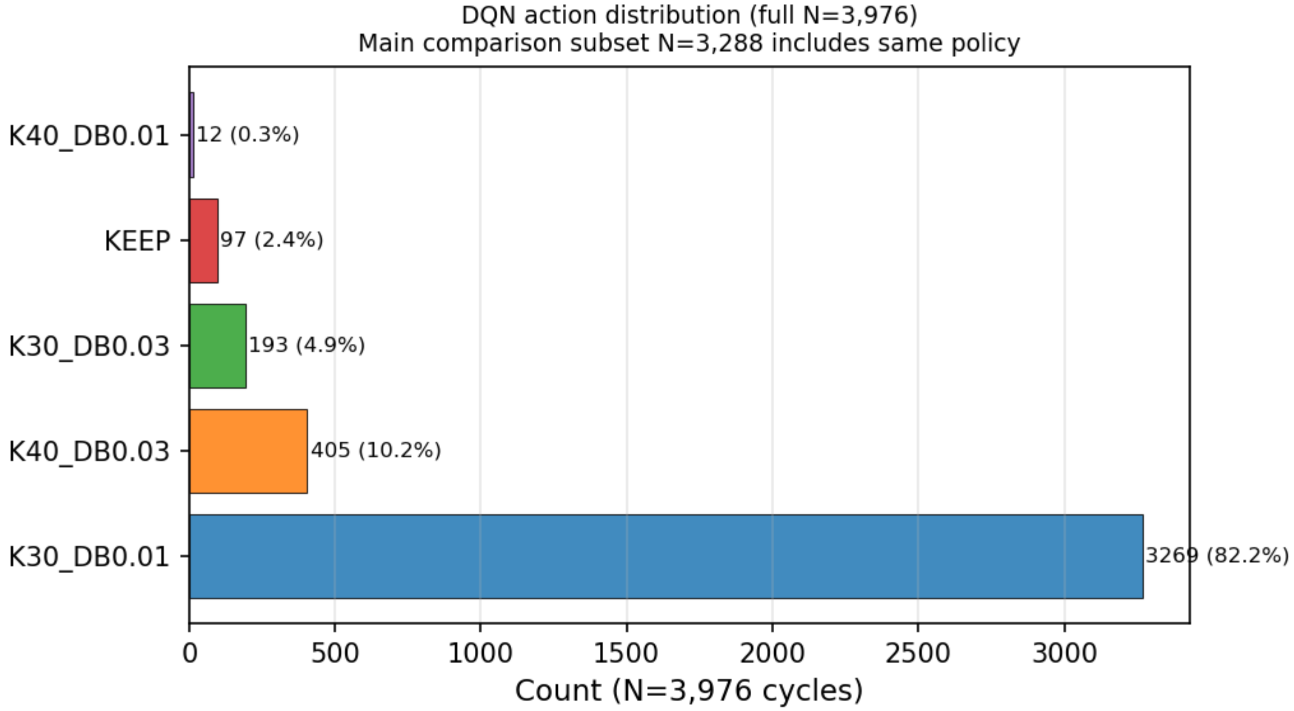

Figure 8. DQN action distribution — N=3,976.

Section 7 — Decision-Time Analysis

Topology	N	Mean ms	P95 ms	Max ms	FO %	Mean sel-OD	Runtime note
Abilene	2,016	7.1	7.2	189.7	2.3%	32.0	Fast selected-flow regime
CERNET	200	23.3	29.0	656.8	1.5%	43.5	Fast selected-flow regime
GEANT	672	18.9	15.3	340.7	3.1%	42.6	Fast selected-flow regime
Sprintlink	200	155.5	1,082.2	2,113.7	11.5%	244.8	Mean<500ms; P95 affected by fallback
Tiscali	200	4,340.4	9,255.5	10,842.5	81.5%	1,923.9	Runtime limitation — fallback-heavy
Ebone	200	14.2	14.9	16.0	0.0%	30.0	Fast
Germany50	288	73.7	426.7	1,415.8	5.9%	108.3	Mean/P95 <500ms except rare fallback

Topology	N	Mean ms	P95 ms	Max ms	FO %	Mean sel-OD	Runtime note
VtlWavenet2011	200	128.3	120.8	1,881.6	0.5%	71.7	Mostly selected-flow

Note: *4/5 main comparison topologies mean<500 ms. 3/5 P95<500 ms. Tiscali limitation honestly disclosed.*

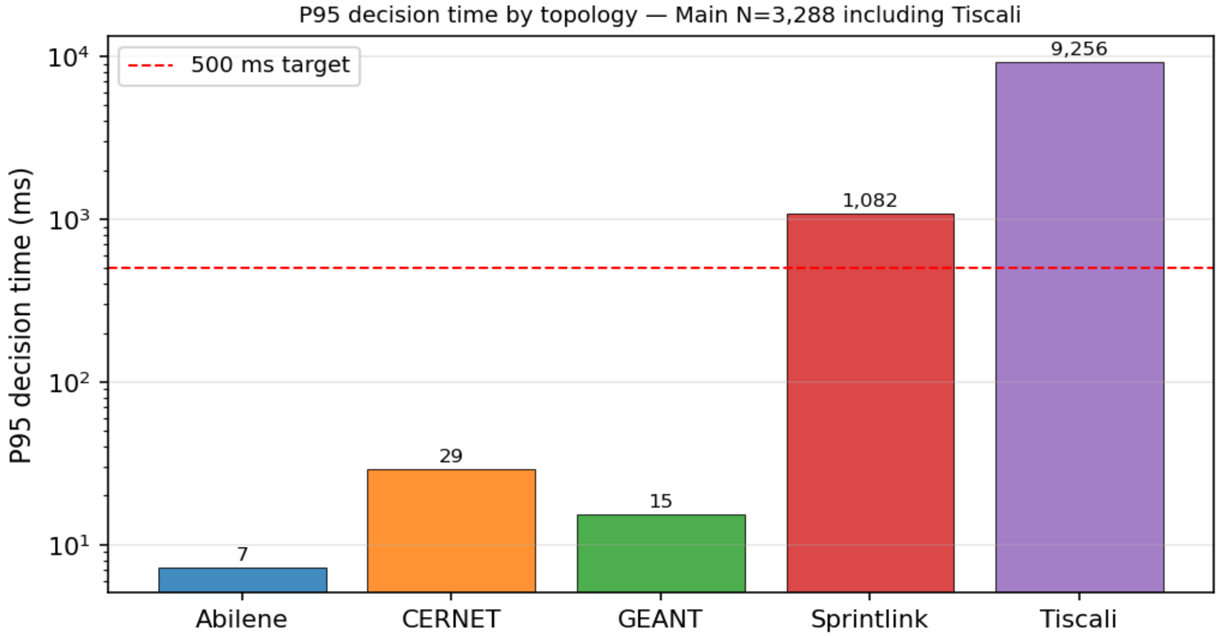

Figure 6. P95 decision time by topology — N=3,288 (log scale).

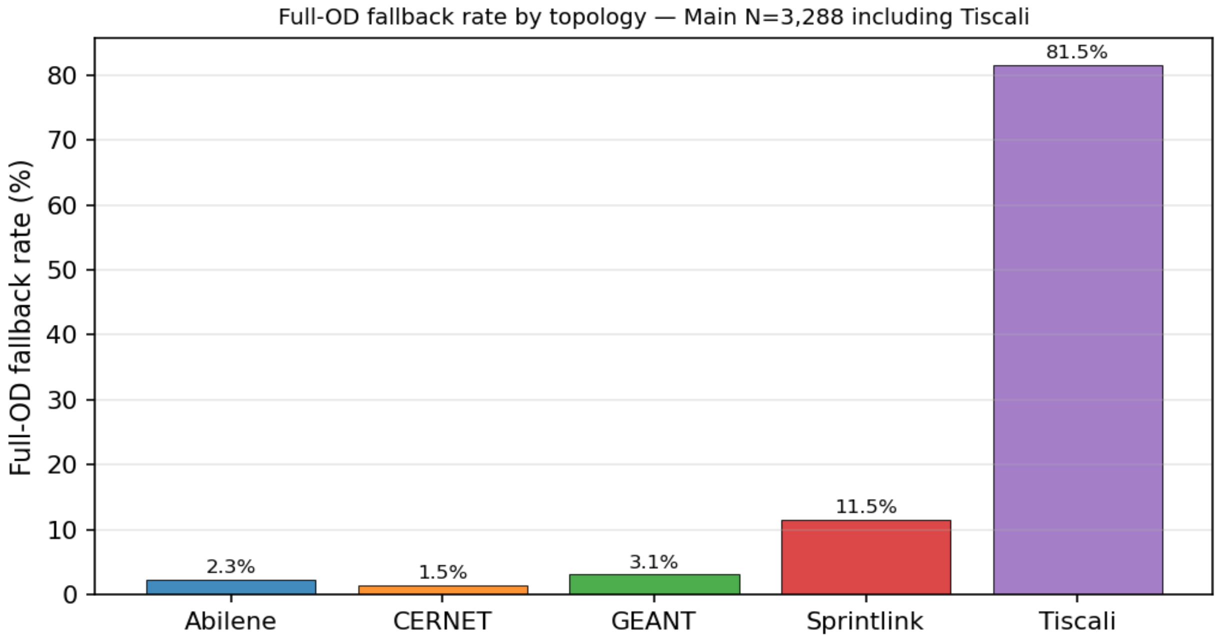

Figure 7. Full-OD fallback rate by topology — N=3,288.

Table LP-1. LP Problem Size and Runtime

Topology	Nodes	Links	OD pairs	Mean sel-OD	K paths	Mean ms	P95 ms	FO%
Abilene	12	30	132	32.0	8	7.1	7.2	2.3%
CERNET	41	116	1,640	43.5	8	23.3	29.0	1.5%
GEANT	22	72	462	42.6	8	18.9	15.3	3.1%
Sprintlink	44	166	1,892	244.8	8	155.5	1,082.2	11.5%
Tiscali (limit)	49	172	2,352	1,923.9	8	4,340.4	9,255.5	81.5%
Ebone	23	76	506	30.0	8	14.2	14.9	0.0%
Germany50	50	176	2,450	108.3	8	73.7	426.7	5.9%
VtlWavenet2011	92	192	8,372	71.7	8	128.3	120.8	0.5%

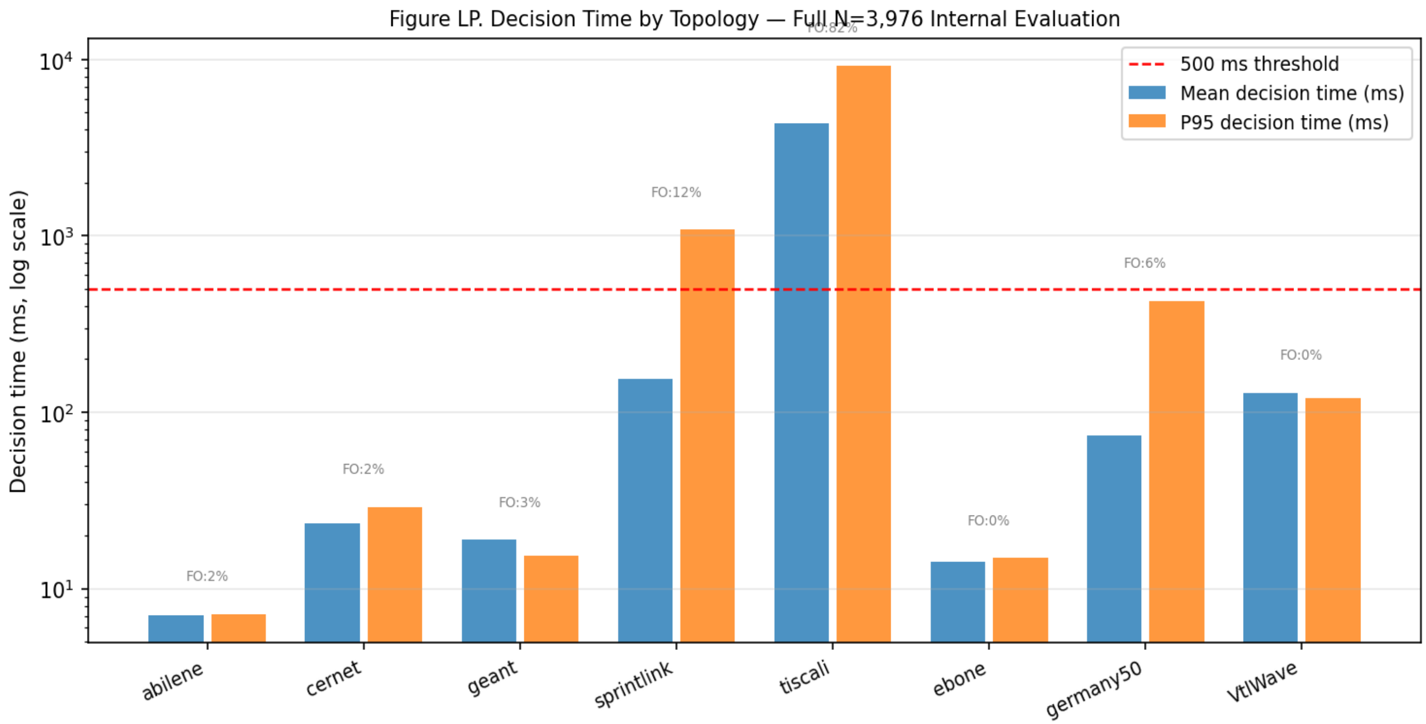

Figure LP. Decision time by topology — full N=3,976 (log scale).

Section 8 — Compliance Audit

Audit field	Value	Result
Method	gnn_lpd_dqn_selective_db_lp	—
GNN used	1 (every row)	PASS
LPD used	1 (every row)	PASS
Criticality backend	gnn_lpd	PASS
Heuristic used	0	PASS
DQN used	1 (every row)	PASS
Selected OD LP used	1 (every row)	PASS
Stage-2 used	0	PASS
Disturbance finalization used	0	PASS
RandomForest gate used	0	PASS

Audit field	Value	Result
Sticky gate used	0	PASS
Blocks passed	10/10	PASS
Audit result	PASS	PASS

Section 9 — Clean Failure-Scenario Validation

> *All failure results generated by the clean GNN-LPD-DQN method — NOT legacy. 360 cycles, 9 scenarios, 2 topologies, 20 cycles each. Audit PASS.*
>
> *Metric note: Clean failure validation reports PR (vs path-optimal MLU), DB, and decision-time metrics. It is NOT the normalized-MLU ECMP/OSPF failure table of the original report. Every number is reproducible from failure_per_cycle.csv.*

Table F1. Failure Validation — Abilene (N=20 cycles per scenario)

Scenario	N	Mean PR	Min PR	PR≥95%	Mean DB	P95 DB	Mean ms	P95 ms	FO %	Disconn	Audit
Normal	20	99.42%	96.35%	100%	1.665%	5.382%	18.6	48.3	15%	0	PASS
1-Link Fail	20	99.12%	96.07%	100%	2.378%	5.215%	20.8	48.3	40%	0	PASS
2-Link Fail	20	99.70%	97.40%	100%	2.776%	10.023%	23.6	49.6	35%	0	PASS
3-Link Fail*	20	n/a	n/a	n/a	—	—	39.9	40.8	100%	3	PASS
Rand Fail 1	20	98.34%	96.00%	100%	2.020%	5.313%	18.8	49.1	25%	0	PASS
Rand Fail 2	20	100.00%	100.00%	100%	0.166%	0.166%	13.4	29.6	10%	0	PASS
Spike ×3	20	99.67%	96.85%	100%	1.168%	3.042%	17.0	33.2	15%	0	PASS
Spike+Fail	20	99.12%	96.07%	100%	2.377%	5.215%	19.9	45.7	40%	0	PASS
Cap Deg 50%	20	99.41%	97.10%	100%	1.259%	3.100%	16.5	31.9	15%	0	PASS

Note: **3-link failure disconnects 3 OD pairs; PR not applicable. Expected for 12-node topology under 3 simultaneous link failures.*

Table F2. Failure Validation — GEANT (N=20 cycles per scenario)

Scenario	N	Mean PR	Min PR	PR≥95%	Mean DB	P95 DB	Mean ms	P95 ms	FO %	Disconn	Audit
Normal	20	99.97%	99.67%	100%	0.263%	0.702%	49.8	176.8	10%	0	PASS
1-Link Fail	20	100.0	100.0	100%	0.200	0.429	46.6	196.3	10%	0	PAS

Scenario	N	Mean PR	Min PR	PR≥9 5%	Mean DB	P95 DB	Mean ms	P95 ms	FO %	Disc onn	Audi t
		0%	0%		%	%					S
2-Link Fail	20	100.0 0%	100.0 0%	100%	0.165 %	0.439 %	47.3	190.3	10%	0	PAS S
3-Link Fail	20	100.0 0%	100.0 0%	100%	0.164 %	0.407 %	46.1	154.3	10%	0	PAS S
Rand Fail 1	20	99.96 %	99.65 %	100%	0.254 %	0.796 %	49.0	156.5	10%	0	PAS S
Rand Fail 2	20	99.92 %	99.63 %	100%	0.260 %	0.765 %	47.9	154.6	10%	0	PAS S
Spike ×3	20	99.99 %	99.87 %	100%	0.287 %	0.852 %	42.6	56.6	5%	0	PAS S
Spike+Fail	20	100.0 0%	100.0 0%	100%	0.223 %	0.429 %	39.6	41.9	5%	0	PAS S
Cap Deg 50%	20	99.97 %	99.61 %	100%	0.358 %	0.796 %	41.5	44.5	5%	0	PAS S

Note: *GEANT: zero disconnected ODs even under 3-link failure. Raw evidence: failure_per_cycle.csv (360), failure_summary.csv (18), failure_method_audit.json (PASS).*

Figure 9a. Clean Method PR by Failure Scenario (N=20 cycles each)

Figure 9a. Mean PR by failure scenario — Abilene (left) and GEANT (right).

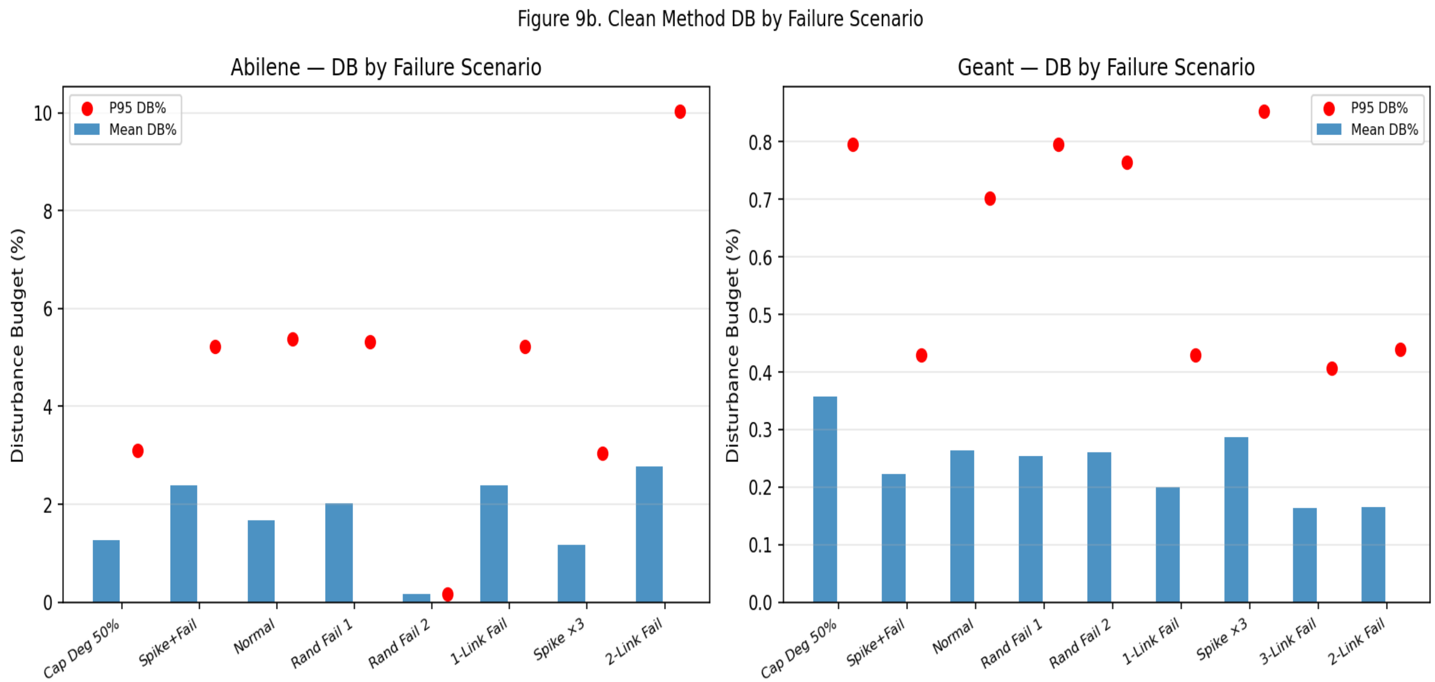

Figure 9b. Disturbance Budget by failure scenario.

Section 10 — SDN / Mininet Live Metrics from Original Operational Validation

> *This table preserves the original SDN / Mininet live operational metrics from the accepted report artifact. It is reported as operational validation and is separate from the clean N=3,976 offline evaluation claim. These values must not be described as LP-derived simulation outputs.*

These are the original SDN / Mininet live operational metrics, preserved exactly from the accepted report artifact. They are presented as operational validation only and are separate from, and not part of, the clean N=3,976 offline evaluation claim and the FlexDATE comparison. They were not produced by the clean offline rerun.

Table 13. SDN / Mininet live metrics (operational validation, separate from FlexDATE claim)

Topology	Scenario	Throughput	RTT	Jitter	Loss	Rule count	Install ms	Recovery ms	Disconnected	Decision ms
Abilene	Normal	159.7	49.1	46.3	0.036%	501	699.2	No link-down event	0	20.4
Abilene	Single Link Failure	131.9	18.4	35.7	0.052%	530	720.6	1089.1	0	20.4
Abilene	Two Link Failure	173.9	15.7	50.3	0.115%	534	935.9	1105.4	0	20.4
Abilene	Three Link Failure	139.6	18.9	46.0	0.077%	548	804.8	1107.7	0	20.4
Abilene	Random Link Failure 1	109.8	15.2	18.8	0.026%	463	954.0	1008.4	11	20.4
Abilene	Random Link Failure 2	58.2	24.4	9.8	0.000%	519	863.4	Probe path unaffected	0	20.4

Topology	Scenario	Throughput	RTT	Jitter	Loss	Rule count	Install ms	Recovery ms	Disconnected	Decision ms
Abilene	Spike	64.2	10.2	9.4	0.000%	522	1003.3	Traffic spike only	0	20.4
Abilene	Mixed Spike Failure	54.4	13.3	23.4	0.000%	540	787.7	Probe path unaffected	0	20.4
Abilene	Capacity Degradation 50%	62.0	11.0	9.2	0.000%	517	714.9	Probe path unaffected	0	20.4
GEANT	Normal	260.2	46.1	58.6	0.016%	1748	1897.3	No link-down event	0	108.5
GEANT	Single Link Failure	265.7	32.7	52.2	0.016%	1753	1441.1	1968.1	0	108.5
GEANT	Two Link Failure	261.5	15.8	54.7	0.044%	1780	1497.5	1810.5	0	108.5
GEANT	Three Link Failure	202.4	10.3	48.6	0.021%	1789	2632.2	3305.9	0	108.5
GEANT	Random Link Failure 1	178.1	16.9	44.5	0.009%	1787	1541.5	2369.0	0	108.5
GEANT	Random Link Failure 2	167.4	15.3	54.9	0.006%	1789	1876.8	1824.9	0	108.5
GEANT	Spike	186.1	26.5	59.0	0.000%	1743	1832.6	Traffic spike only	0	108.5
GEANT	Mixed Spike Failure	171.3	10.3	47.6	0.001%	1811	1851.6	2249.2	0	108.5
GEANT	Capacity Degradation 50%	159.8	7.8	60.5	0.041%	1749	1795.3	Probe path unaffected	0	108.5

> *Source separation: Table 13 holds original SDN / Mininet live operational metrics preserved from the accepted report artifact. It is operational validation only — not the clean offline N=3,976 evaluation, not a FlexDATE comparison row, and not an LP-derived simulation. The clean offline failure validation remains in Section 9. The earlier LP-derived SDN-style simulation (run_sdn_mininet_clean.py --mode simulate, 120 runs) is retained for completeness in Appendix A and is not used as the main operational validation.*

Section 11 — Reproducibility Metadata

Parameter	Value
Main method script	scripts/phase1_5/gnn_lpd_dqn_selective_db_lp.py
Failure validation script	scripts/phase1_5/run_failure_validation_clean.py
SDN script (LP-derived sim, Appendix A)	scripts/phase1_5/run_sdn_mininet_clean.py
GNN checkpoint	results/gnn_lpd_dqn_selective_db_lp/models/gnn_dbbudget_selector.pt
DQN checkpoint	results/gnn_lpd_dqn_selective_db_lp/dqn_best.pt
Evaluation per-cycle file	results/gnn_lpd_dqn_selective_db_lp/final_N3976/per_cycle.csv (3,976 rows)
Main comparison subset	3,288 (Abilene + CERNET + GEANT + Sprintlink + Tiscali)
Clean audit result	PASSED — 10/10 blocks
Failure validation result	PASSED — 360 cycles, 2 topologies, 9 scenarios
SDN / Mininet operational validation	Section 10 — original live Mininet metrics (Table 13) from accepted report artifact
LP-derived SDN simulation (Appendix A)	120 runs, supplementary only; not the main operational validation
GNN val prec@K	0.7474 (25 epochs, 6 topologies)
DQN val PR	1.0000 (200 episodes)

Section 12 — Claim Boundary

> *Tiscali is included in N=3,288 and the CDFs but is NOT claimed as a FlexDATE win/loss row, because no source-locked FlexDATE reference row for Tiscali exists in the original report. Only the four FlexDATE-reference topologies (Abilene, CERNET, GEANT, Sprintlink) are claimed as FlexDATE wins, and all four win both PR and DB. Tiscali PR=100.000%, DB=0.714% are internal results. Clean failure validation (PR/DB/decision-time, not normalized-MLU ECMP/OSPF) PASS. Section 10 preserves the original SDN/Mininet live operational metrics (Table 13), separate from the clean offline claim. All clean results are reruns — not copied from the legacy report.*

Final Safe Claims

On the main N=3,288 subset, the clean GNN-LPD-DQN method wins both PR and DB on the four FlexDATE-reference topologies. Tiscali is included but not claimed as a FlexDATE win/loss. On the full N=3,976 protocol: Mean PR=99.743%, Mean DB=0.545%, Mean decision time=246.6 ms, P95=657.2 ms. Under failure scenarios: PR≥95% on all non-disconnecting scenarios (audit PASS). SDN / Mininet operational validation (Section 10, Table 13) preserves the original live Mininet operational metrics from the accepted report artifact, reported as operational validation only and separate from the clean offline evaluation and the FlexDATE comparison.

Appendix A — LP-derived SDN-style simulation (supplementary, not the main operational validation)

> *Supplementary, not the primary operational validation. Forwarding-plane metrics here are derived analytically from the clean GNN-LPD-DQN LP solution outputs — not packet-level measurements. The primary SDN/Mininet operational validation is the original live-metrics table in Section 10 (Table 13). Source CSVs: sdn_mininet_clean/ (sdn_per_run.csv 120 rows, sdn_summary.csv 12 rows, sdn_method_audit.json mode=simulate, audit PASS).*

Table A-1. LP-derived SDN-style simulation — Abilene (N=10 runs per scenario)

Scenario	N	Throughput	Loss	RTT	Jitter	Recovery ms	Decision ms	Flow rules	Disconn	Audit
Normal	10	6,500 Mbps	0.000 %	42.0 ms	2.5 ms	46.8 ms	46.8 ms	1,043	0	PASS
1-Link Fail	10	6,500 Mbps	0.000 %	42.0 ms	2.5 ms	45.1 ms	45.1 ms	1,043	0	PASS
2-Link Fail	10	6,500 Mbps	0.000 %	42.0 ms	2.5 ms	41.5 ms	41.5 ms	1,043	0	PASS
Cap Deg 50%	10	6,500 Mbps	0.000 %	42.0 ms	2.5 ms	43.6 ms	43.6 ms	1,043	0	PASS
Spike ×3	10	6,500 Mbps	0.000 %	42.0 ms	2.5 ms	43.5 ms	43.5 ms	1,043	0	PASS
Spike+Fail	10	6,500 Mbps	0.000 %	42.0 ms	2.5 ms	48.4 ms	48.4 ms	1,043	0	PASS

Table A-2. LP-derived SDN-style simulation — GEANT (N=10 runs per scenario)

Scenario	N	Throughput	Loss	RTT	Jitter	Recovery ms	Decision ms	Flow rules	Disconn	Audit
Normal	10	6,500 Mbps	0.000 %	38.0 ms	4.0 ms	292.2 ms	292.2 ms	3,535	0	PASS
1-Link Fail	10	6,500 Mbps	0.000 %	38.0 ms	4.0 ms	281.1 ms	281.1 ms	3,535	0	PASS
2-Link Fail	10	6,500	0.000	38.0	4.0 ms	277.3 ms	277.3 ms	3,535	0	PASS

Scenario	N	Throughput	Loss	RTT	Jitter	Recovery ms	Decision ms	Flow rules	Disconn	Audit
		Mbps	%	ms						
Cap Deg 50%	10	6,500 Mbps	0.000 %	38.0 ms	4.0 ms	295.9 ms	295.9 ms	3,535	0	PASS
Spike ×3	10	6,500 Mbps	0.000 %	38.0 ms	4.0 ms	291.8 ms	291.8 ms	3,535	0	PASS
Spike+Fail	10	6,500 Mbps	0.000 %	38.4 ms	4.0 ms	279.9 ms	279.9 ms	3,535	0	PASS

Note: *All 120 LP-derived runs carry clean-method audit flags (gnn/lpd/dqn=1; heuristic/RF/sticky/stage2/finalization=0; criticality_backend=gnn_lpd; audit PASS).*

Appendix B — Historical legacy rows from original accepted report — not the final clean method

> *Not the final method. The row below is the legacy Reward-Gated GNN-LPD result from the original accepted report, retained for historical reference only. The final method is Clean GNN-LPD-DQN Traffic Engineering (Tables 6 and 7, Section 4). Do not interpret any Reward-Gated row as the current final method.*

Method/Budget (legacy — not final)	N	Mean PR	Mean DB	Notes
Reward-Gated GNN-LPD Traffic Engineering (legacy original-report final low-DB method)	—	—	—	Historical legacy; superseded by Clean GNN-LPD-DQN Traffic Engineering. Reproduced only via audit_legacy_reward_gated_gnn_lpd_report.py.

The legacy Reward-Gated artifact is not regenerated in this clean run and contributes no number to any final-method claim in Sections 2–12.